Networking Academy

Altea Kara A. Blaza
CPE11S4

Lab - Investigate BIOS or UEFI Settings

Introduction

In this lab, you will boot the computer, explore the firmware setup utility program, and change the boot order sequence.

Recommended Equipment

- Computer with or without operating system
- Motherboard manual

Instructions

Part 1: Enter BIOS or UEFI.

Step 1: Power on the computer.

a. Plug the power supply cable into an AC wall outlet.

b. If there is a power switch on the power supply, set the switch to “1” or “on”.

c. Turn on the computer with the power button on the front panel.

Note: If the computer beeps more than once, or if the power does not come on, notify your instructor.

Step 2: Enter the firmware setup program.

During POST, press the firmware setup key or key combination. The firmware setup utility program screen will appear.

Questions:

What is the key or combination of keys used to enter the firmware setup utility program?

I’ll see a brief message on the splash screen that says something like 'Press F2 to enter Setup'. Also, sometimes it’s F10 or Esc depending on the brand of the motherboard.

Who manufactures the BIOS or UEFI system for your computer?

I can usually find the manufacturer’s name right at the top of the screen or under a 'Main' or 'System Information' tab. The big names I usually see are American Megatrends (AMI), Phoenix Technologies, or Insyde. If I’m on a branded laptop like a Dell or HP, it might just show their own logo as the firmware provider.

What is the BIOS or UEFI version?

I'd find the version number listed in that same 'System Information' section. It might say something like 'Version 2.17.1246' or a shorter code like 'A05'. And knowing this is really important if I ever need to update the firmware to support new hardware or fix security bugs.

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Part 2: Explore the Settings.

Step 1: List the main menu options.

Question:

List the main menu options and describe what is monitored in each menu?



The Main is where I go for the basic overview. It monitors the system date and time, and it's usually where I verify the hardware basics, like the BIOS version, what kind of processor is installed, and exactly how much RAM the system is detecting.

In the Advanced menu, this section is used to configure specific hardware settings, like enabling or disabling onboard devices (such as the audio or LAN), adjusting SATA storage modes, and fine-tuning advanced CPU features.

The Power tab is all about energy management. I'd use this to monitor and configure how the computer handles sleep modes, power-saving features, and what happens when the power button is pressed.

This is the Boot menu, which is where I control the startup sequence. This is where I'd go to set the boot priority, telling the computer to look at the optical drive or a USB stick before the hard drive, and where I can toggle settings like 'Fast Boot' or 'Secure Boot'.

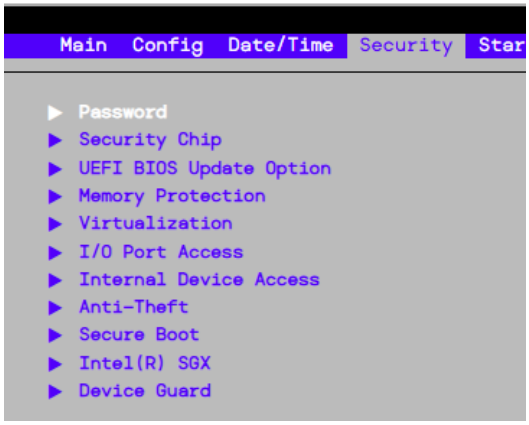
Exit tab is for wrapping things up. It gives me the option to save all the changes I've made and restart the computer, or if I think I made a mistake, I can choose to discard the changes and exit without saving. It's also where I can reset everything back to the factory default settings if needed.

Step 2: Find the security settings.

Navigate through each screen to find the security settings.

Question:

What security settings and features are available?



According to the given image, these are the features that are available in the security settings.

Step 3: Find the CPU settings.

Navigate through each screen to find the CPU settings.

Questions:

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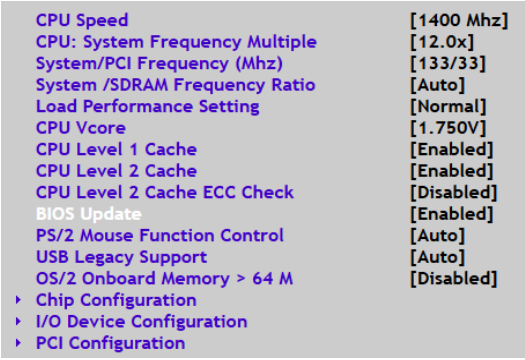
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What is the CPU speed?



The measure of how many cycles a processor can run per second.

What other information is listed for the CPU?



According to the given image, these are the features that are available in the CPU settings.

Step 4: Find the RAM settings.

Navigate through each screen to find the RAM settings.

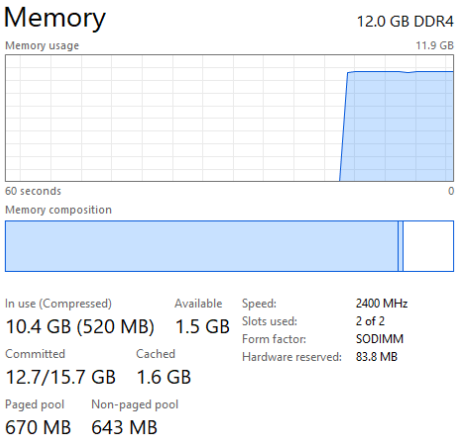
Questions:

What is the RAM speed?



It is the measure of how fast the computer's memory can read and write data.

What other information is listed for the RAM?



According to the given image, these are the features that are available in the RAM settings.

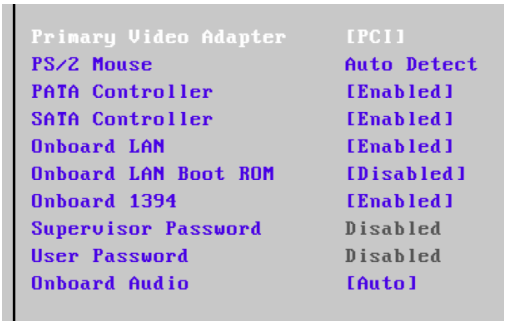
Step 5: Find the hard drive settings.

Navigate through each screen to find the hard drive settings.

Questions:

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What information is listed for the hard drive?



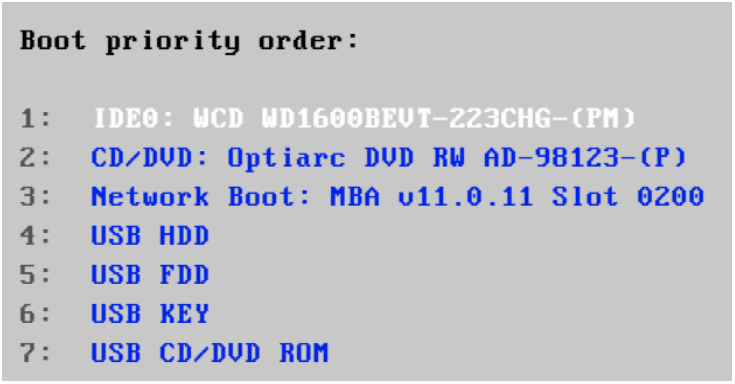
According to the given image, these are the features that are available in the Hard Drive settings.

Step 6: Find the boot order sequence.

Navigate through each screen to find the boot order sequence.

Questions:

What is the first boot device in the boot order sequence?



Right now, the first device in the sequence is the IDE0: WCD WD1600BEVT-223CHG-(PM). Essentially, that's the internal Western Digital hard drive where the main operating system is stored.

How many additional devices can be assigned in the boot order sequence?

According to the image there are six additional devices that can be assigned to follow it in the boot priority.

Step 7: Set the device boot order settings.

a. Ensure that the first boot order device is the optical drive.

b. Ensure that the second boot order device is the hard disk drive.

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Questions:

Why would you change the first boot device to the optical drive?

I'd set the optical drive as the first priority primarily because I need the computer to look for external software before it loads the operating system on the hard drive. This is what I'd do if I were trying to install a new version of Windows from a DVD, or if I needed to run a system recovery disc or a diagnostic tool to fix a problem.

What happens when the computer boots and the optical drive does not contain bootable media?

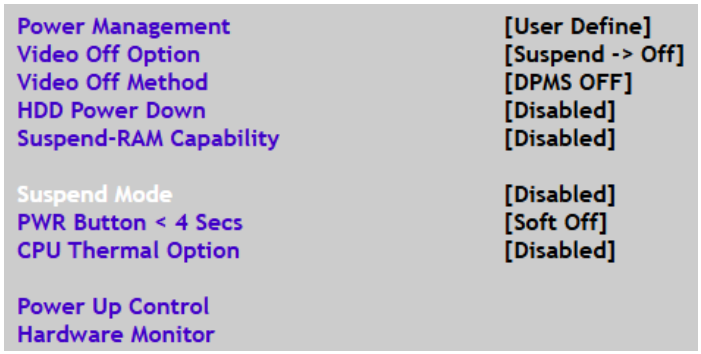
If there's no disc in the tray or if there is a disc but it doesn't have those specific 'boot' files, the computer doesn't just stop or crash. It simply moves down to the next item on my priority list. Since I set the hard drive as the second device, the BIOS will automatically skip the optical drive and boot up the operating system from the hard drive like it normally does. It adds a few seconds to the startup process, but otherwise, everything runs as usual.

Step 8: Find the power management setup or ACPI screen.

Navigate through each screen to find the power management setup screen, or ACPI screen.

Question:

What power management settings are available?



According to the given image, these are the features that are available in the Power management settings.

Step 9: Find the PnP settings.

Navigate through each screen to find the PnP settings.

Question:

What PnP settings are available?



According to the given image, this is the only feature that is available in the PnP settings.

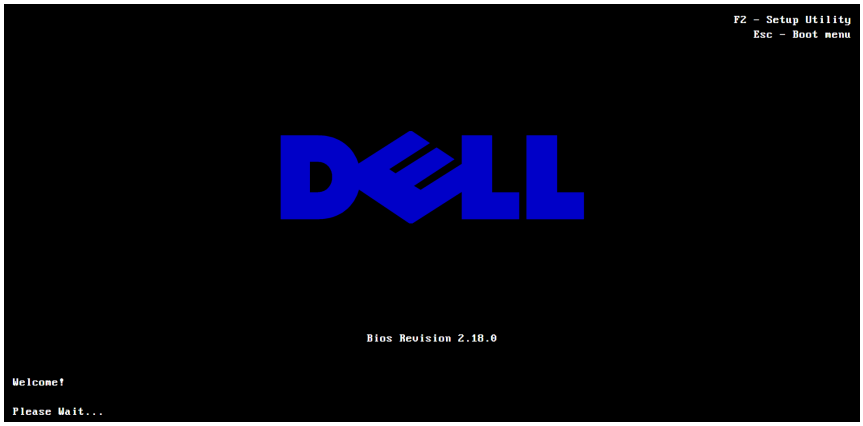
Step 10: Find the splash screen settings.

Navigate through each screen to find the splash screen settings.

Question:

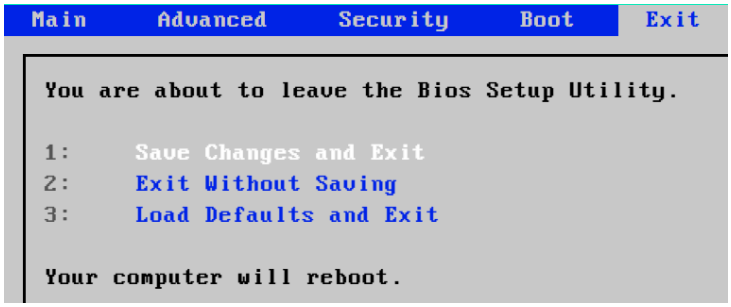
What splash screen settings are available?

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Step 11: Save and exit the setup utility program.

Save the new BIOS/UEFI settings and exit the setup utility program. The computer should restart automatically.



Note: An error message stating that an OS cannot be found (or a similar error) will appear on the screen after the computer boots. An operating system must now be installed to prevent this error. It is safe to turn off the computer at this time.

This lab is complete. Please have the instructor verify your work.

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